Kartavya Soni

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SUMMARY

Data Scientist with 3 years of experience solving insurance and financial-services problems through predictive modeling, statistical analysis, and responsible AI. Developed solutions for claims routing, fraud detection, credit-risk assessment, financial document retrieval, and model monitoring, with a focus on reliable evaluation and explainable decisions. Targeting Data Scientist roles where I can build practical machine learning and Generative AI solutions that improve operational decisions and customer outcomes.

SKILLS

Programming & Query Development: Python, SQL, Data Structures, Complex Queries, Joins, Common Table Expressions, Window Functions

Data Analysis & Feature Engineering: Pandas, NumPy, PySpark, Exploratory Data Analysis, Data Cleaning, Feature Engineering, Missing-Value Treatment, Outlier Analysis, Statistical Analysis, Data Validation

Machine Learning & Predictive Modeling: Scikit-learn, XGBoost, LightGBM, Random Forest, Logistic Regression, Classification, Anomaly Detection, Fraud Detection, Credit Risk Modeling, Optuna, GridSearchCV

Generative AI & LLM Engineering: Azure OpenAI, Large Language Models, Retrieval-Augmented Generation, Prompt Engineering, Azure AI Search, Vector Embeddings, Agentic Workflows, Multimodal AI, Structured Outputs

Cloud, Big Data & AI Platforms: Microsoft Azure, Azure Machine Learning, Azure AI Foundry, Azure Databricks, Azure AI Vision, AWS S3, AWS EC2, Apache Spark

MLOps, Deployment & Automation: MLflow, Model Registry, Apache Airflow, Docker, Kubernetes, Flask, REST APIs, CI/CD Pipelines, Jenkins, Terraform, Infrastructure as Code

Model Evaluation, Monitoring & Responsible AI: SHAP, Cross-Validation, Model Calibration, Precision, Recall, F1-Score, ROC-AUC, PR-AUC, Data Drift, Model Drift, Fairness Testing, Human-in-the-Loop Validation, AI Guardrails, PII Protection

Databases, Visualization & Development Tools: PostgreSQL, Power BI, Matplotlib, Streamlit, Git, Model Performance Dashboards, Stakeholder Reporting

EXPERIENCE

The Allstate Corporation, USA | AI Data Scientist — Feb 2026 – Present

- Built claims-reassignment training data from policy, loss, and customer-contact systems at ~150K claims/month, using only fields available at reassignment time to prevent leakage from downstream repair and payment records.
- Engineered severity, documentation, dispute, and workload features, improving macro-averaged routing recall by 12% relative to the existing rules-based baseline on an out-of-time holdout.
- Trained XGBoost and LightGBM models, using SHAP to explain route recommendations and identify claim characteristics influencing adjuster assignment decisions during validation reviews.
- Built an Azure OpenAI retrieval prototype that summarized claim notes and surfaced policy guidance, reducing median manual review time by 18% versus the existing workflow.
- Created human-reviewed communication drafts with prompt templates, factual checks, and PII masking, improving consistency while keeping adjusters responsible for final customer messages.
- Supported model validation and experiment tracking in Azure Machine Learning and MLflow, documenting performance, limitations, and reviewer approvals before controlled pilot testing.

Mphasis, India | Data Scientist — Jan 2022 – Jun 2024

- Built transaction-monitoring datasets from card payments, merchant activity, device signals, and account histories using PySpark and Spark SQL for fraud analysis.
- Investigated suspicious behavior through velocity checks, merchant clustering, and sequence analysis, helping analysts identify coordinated fraud patterns and emerging payment risks.
- Prevented data leakage by excluding post-investigation fields, using out-of-time validation, and fitting preprocessing transformations only within cross-validation folds during model training.
- Improved confirmed-fraud recall by 16% relative to the legacy rules baseline using Random Forest, Isolation Forest, class weighting, and threshold optimization.
- Reduced false-positive alerts by 13% relative to the prior monitoring workflow, measured across three monthly review cycles using segment-specific decision thresholds.
- Developed Power BI dashboards showing alert volumes, fraud trends, investigator outcomes, and model stability, supporting weekly reviews with risk and operations teams.
- Operationalized batch scoring through Airflow, Docker, AWS, and REST APIs, enabling scheduled fraud detection, versioned releases, and controlled production support.

EDUCATION

Master of Science in Computer Science **May 2026**
Florida Atlantic University, USA GPA: 3.96
Bachelor of Engineering in Computer Engineering **May 2024**
Gujarat Technological University, India GPA: 3.70

CERTIFICATIONS

Google Advanced Data Analytics Professional Certificate – Google
AWS Certified Cloud Practitioner – AWS

PROJECTS

FilingQA: RAG Analyst for SEC Filings | *FastAPI, pgvector, Claude API, Sentence Transformers, SEC EDGAR*
More Info

- Built a RAG research assistant that ingested SEC 10-K filings from EDGAR, segmented documents by section, and prepared searchable financial-content chunks.
- Implemented semantic retrieval with Sentence Transformers and pgvector, then exposed ranked passages through FastAPI endpoints for responsive analyst questions and downstream applications.
- Generated citation-grounded answers with Claude API and controlled prompts, reducing unsupported interpretations by requiring every response to reference retrieved filing passages.
- Created a 100-question evaluation set measuring hit-rate@5, retrieval relevance, citation coverage, and answer faithfulness before releasing the application for portfolio demonstration.

Global Lens: Global Economic Progress Analysis | *Python, FastAPI, Scikit-learn, ARIMA, Streamlit, Docker*
More Info

- Developed a Python and FastAPI ingestion pipeline for 1,000+ global indicators, adding response caching that reduced median API latency by 60% versus uncached requests.
- Applied PCA and K-Means to group countries into five economic profiles, enabling comparisons across regions, income levels, and long-term development patterns.
- Trained ARIMA models with rolling-window validation, reducing GDP forecast MAPE by 12% relative to a historical-average baseline across evaluated periods.
- Deployed a Dockerized Streamlit dashboard with scenario controls, allowing users to compare countries, indicators, forecasts, and cluster profiles through an interactive interface.

Loan Meter: Financial Risk Prediction Engine | *Python, Pandas, XGBoost, Bayesian Optimization, SHAP*
More Info

- Processed 200,000+ loan records through Pandas pipelines, applying missing-value treatment, outlier controls, feature validation, and reproducible preprocessing before model development.
- Optimized an XGBoost credit-risk classifier through Bayesian search, achieving 0.76 ROC-AUC on held-out data and improving separation of higher-risk applicants.
- Applied cost-sensitive learning to address a 10:1 class imbalance, increasing minority-class recall by 14% relative to the standard-threshold baseline while preserving approval precision.
- Generated SHAP-based reason codes for individual predictions, helping reviewers understand default drivers and supporting transparent model documentation for compliance and audit discussions.